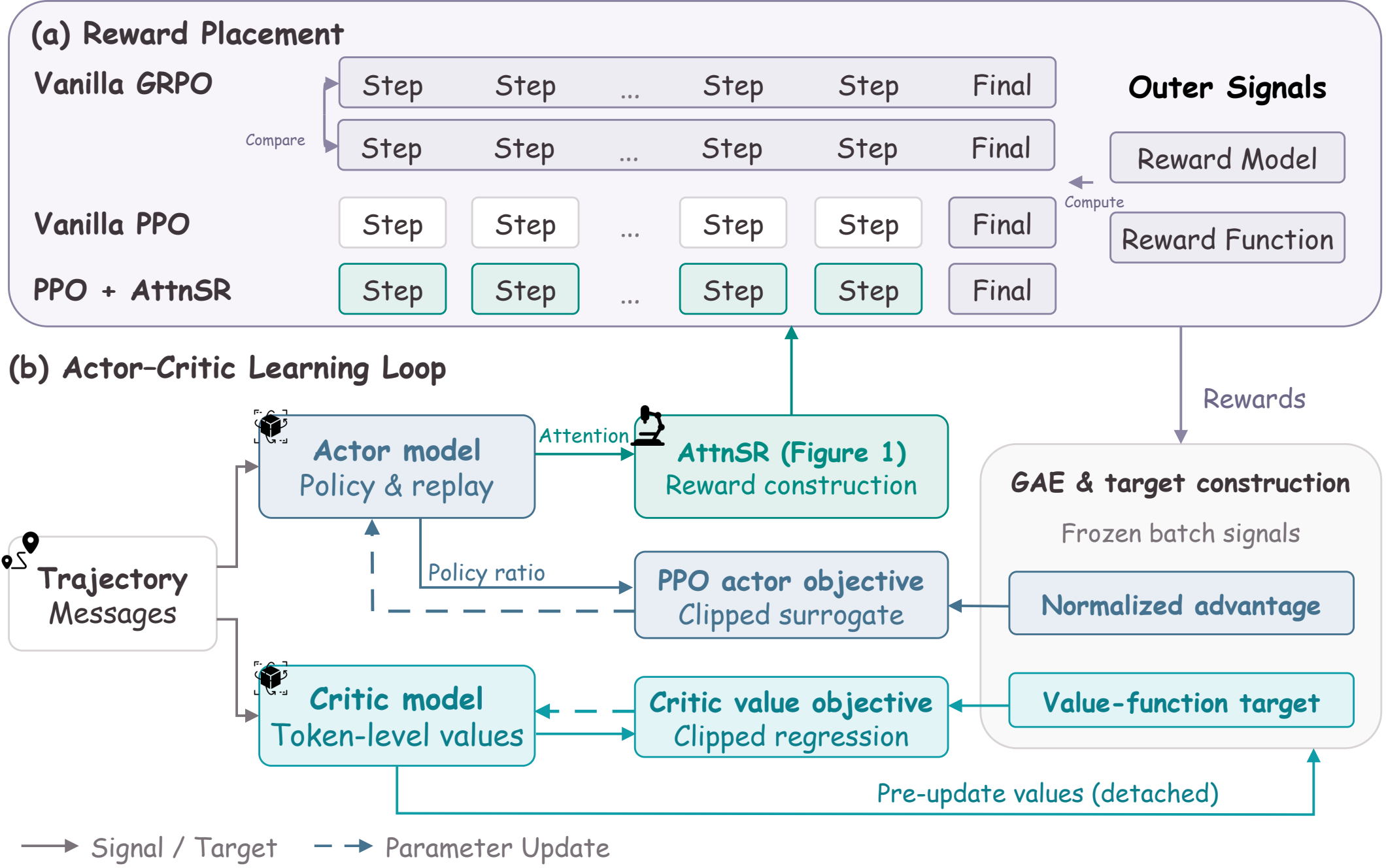

(a) Reward Placement
Vanilla GRPO
Step Step ... Step Step Final
Compare
Step Step ... Step Step Final
Outer Signals
Reward Model
Compute
Reward Function
Vanilla PPO
Step Step ... Step Step Final
PPO + AttnSR
Step Step ... Step Step Final
(b) Actor–Critic Learning Loop
Rewards
Actor model
Policy & replay
Attention
AttnSR (Figure 1)
Reward construction
GAE & target construction
Frozen batch signals
Trajectory
Messages
Policy ratio
PPO actor objective
Clipped surrogate
Normalized advantage
Critic model
Token-level values
Critic value objective
Clipped regression
Value-function target
Pre-update values (detached)
Signal / Target
Parameter Update